

Task 4: SQL for Data Analysis

Objective: Use SQL queries to extract and analyze data from a database.

Tools: MySQL or PostgreSQL or SQLite

Deliverables: SQL queries in a SQL file + screenshots of output

Hints/Mini Guide:

- a. Use SELECT, WHERE, ORDER BY, GROUP BY
- b. Use JOINS (INNER, LEFT, RIGHT)
- c. Write subqueries
- d. Use aggregate functions (SUM, AVG)
- e. Create views for analysis
- f. Optimize queries with indexes

Dataset: Ecommerce_SQL_Database(or any data set of your choice)

Outcome: Learn to manipulate and query structured data using SQL.

Interview Questions:

1. What is the difference between WHERE and HAVING?
2. What are the different types of joins?
3. How do you calculate average revenue per user in SQL?
4. What are subqueries?
5. How do you optimize a SQL query?
6. What is a view in SQL?
7. How would you handle null values in SQL?

Task Submission Guidelines

- Time Window:

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- Self-Research Allowed:

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- Debug Yourself:

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- No Paid Tools:

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- GitHub Submission:

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.